Project-Based Learning (PBL) – Individual Project Guidelines

Objective

The objective of this PBL activity is to provide students with hands-on experience in software development, project management, implementation, and testing using JIRA.

Each student is required to work individually on a software project and implement and test any two modules of the proposed system.

Project Requirements

Each student should:

1. Select any two modules for implementation from the problem statement.
2. Create and manage the project using JIRA.
3. Design and execute test cases for the implemented modules.
4. Record and track identified defects using JIRA.
5. Demonstrate the working modules and testing results.

JIRA Activities

Each student must create an individual JIRA project and use it for managing the project.
The JIRA project should include:

- Project name and description
- Project requirements
- Epics for major functionalities
- User Stories
- Tasks and Sub-tasks
- Sprint planning
- Task assignment
- Task status updates
- Bug/Issue tracking
- Sprint progress

Implementation

Each student must implement any two functional modules of the selected system.
For each module, students should provide:

- Module description
- Functional requirements
- User interface
- Database/backend functionality, wherever applicable
- Input validation
- Working implementation
- Integration with other components, if required

Testing

Students must test both implemented modules.

Test Cases

Prepare test cases using the following format:

Test Case ID	Module	Test Scenario	Test Data	Expected Result	Actual Result	Status

Students should apply appropriate testing techniques such as:

- Functional Testing
- Unit Testing
- Integration Testing
- Boundary Value Analysis
- Equivalence Partitioning
- Validation Testing

Bug Tracking Using JIRA

Any defects identified during testing should be created as issues/bugs in JIRA.

Each bug should contain:

- Bug ID
- Bug description
- Steps to reproduce
- Severity/Priority
- Status
- Resolution

Final Demonstration

During the PBL evaluation, each student should individually demonstrate:

1. Project title and problem statement
2. System/module description
3. JIRA project
4. Product backlog
5. User stories
6. Implementation of any two modules
7. Source code
8. Test cases and test results
9. Bugs/issues recorded in JIRA
10. Bug-fixing and retesting evidence
11. Final working demonstration
12. Project report and presentation

Evaluation Rubrics for PBL

Sr. No.	Evaluation Component	Criteria	Marks
1	**JIRA Project Management**	• Project creation, requirements, epics, user stories, tasks/sub-tasks, sprint planning, and status updates • Identification, documentation, tracking, fixing, and retesting of defects using JIRA	5
3	**Testing & Test Cases**	• Functionality, completeness, UI, database/backend, validation, and integration of the two selected modules • Preparation and execution of test cases; appropriate use of testing techniques; correctness of expected/actual results	5
	Total		**10**